INTERNSHIP PROJECT REPORT

Diya

An unattended meditation kiosk

PRESENTED BY

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UNDER THE GUIDANCE OF

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Mentor

INTRODUCTION

What I worked on

A visitor walks up to a kiosk with nobody there to help them. My work was everything around the experience: identifying the visitor, running their session, and handing back a report — then resetting for the next person.

01

The kiosk application

A fullscreen desktop app in C# and .NET with Avalonia, running on Linux.

02

Phone-based identification

A backend and web pages so a visitor is identified using only their own phone.

03

Hardware integration

Connecting to the other team's application without touching their code.

04

Packaging and deployment

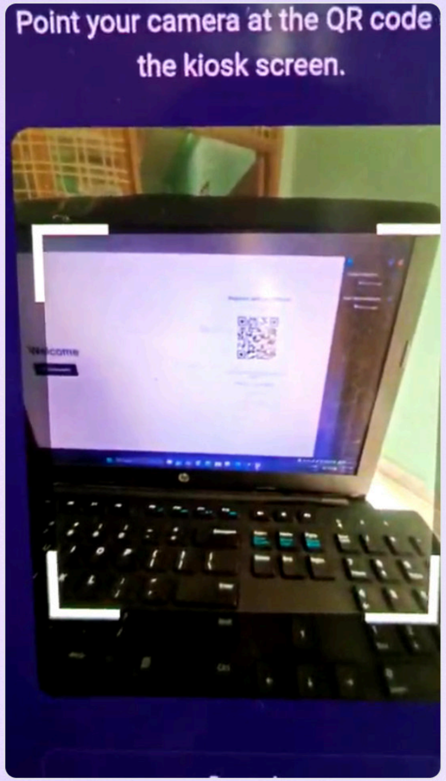
A single installable package that boots straight into the kiosk and restarts itself.

THE QR CODE SYSTEM

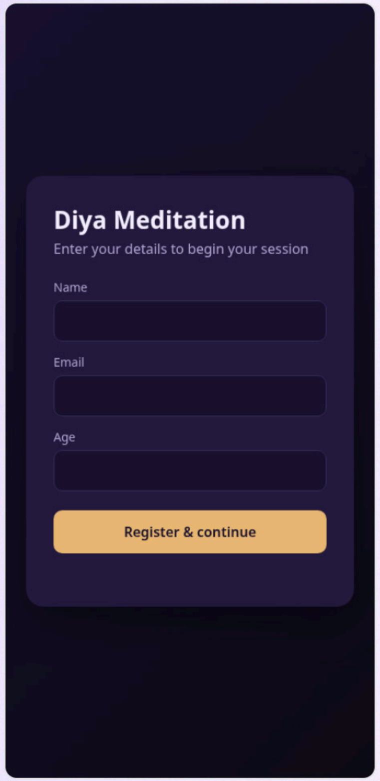
The kiosk shows the code. The phone reads it.

There was no scanner hardware, so I inverted it. The kiosk asks the backend for a session and displays it as a QR code. The visitor's phone scans that code, and the kiosk — which is polling the backend — moves ahead on its own.

- No scanner, no app install, no typing
- The phone is the camera and the keyboard
- Phone and kiosk never talk directly — only through the backend



The scan, filmed on the real setup — plays in full



Scanning the code opens this page on the visitor's phone

END TO END

The whole flow

01

Identify

The visitor opens their private link and scans the kiosk with their phone.

02

Claim

The backend ties that person to the open kiosk session. The kiosk sees it while polling.

03

Session

The kiosk starts the camera pipeline and waits while the meditation session runs.

04

Report

The generated PDF is found and shown on screen, then the kiosk resets itself.

> The visitor never presses a button to begin — being identified is what starts the session.

PROBLEMS I HAD TO SOLVE

Two real problems

`Problem 01`

Identifying someone with no scanner

Every obvious answer needed hardware — a QR reader, a card reader, or a keyboard strangers would share. I tried putting the visitor's data in the QR code, then putting an ID in it, and both still needed a reader.

APPROACH

Turn it around: the kiosk displays the QR and the visitor's phone does the scanning. The constraint made the design simpler.

`Problem 02`

Talking to another team's app without changing it

The meditation software shipped as a separate `.deb` application. It had no API to call, and I could not modify it — it kept changing on their side all summer.

APPROACH

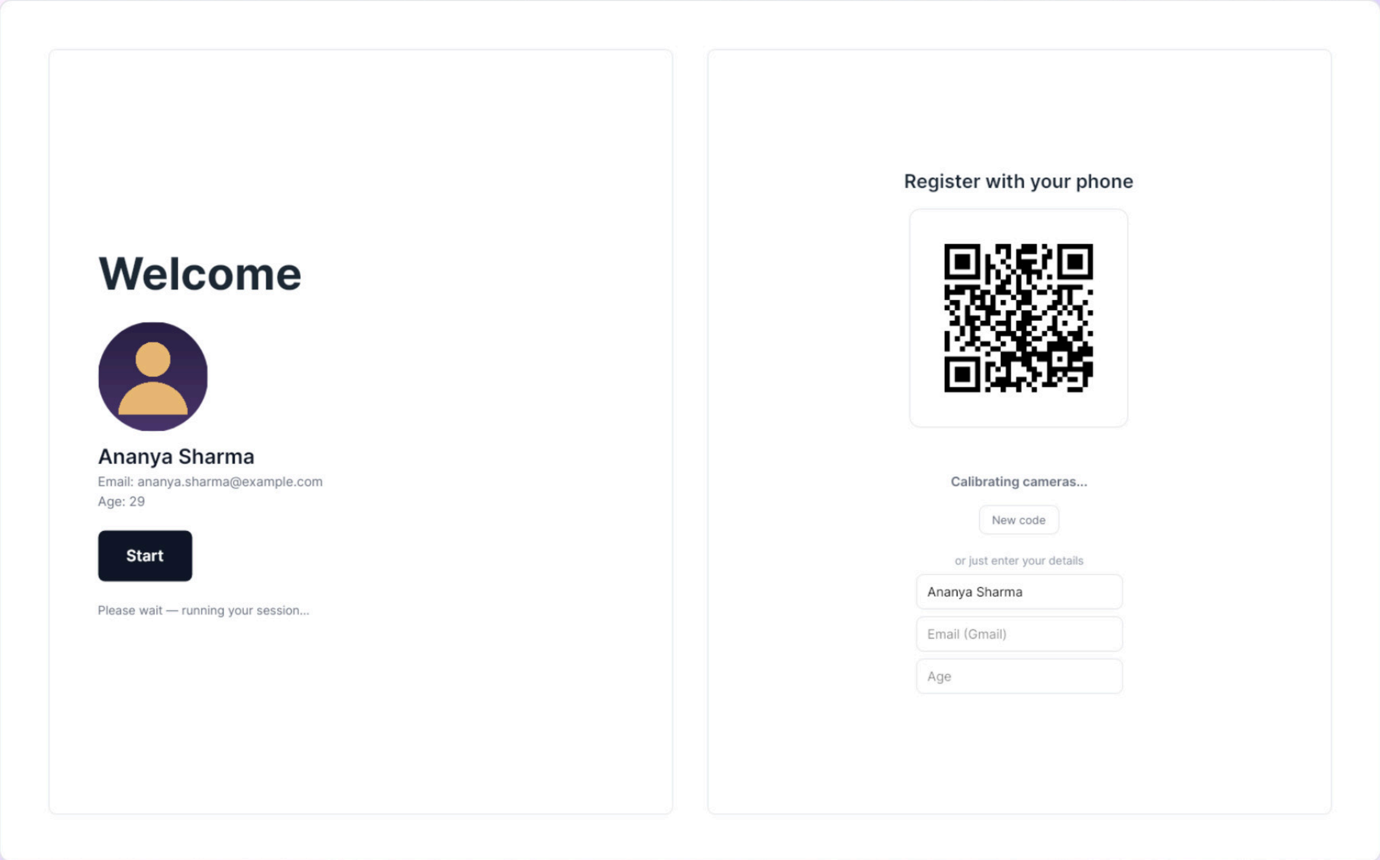
My app launches their program and then simply watches the folder. It keeps checking whether the report PDF has been created, and only accepts a file newer than the moment the session started — so nobody gets the previous visitor's report. Once the file appears, the kiosk renders it.

> One process to launch and one file to watch — so their internals could change freely without ever breaking my application.

SEE IT RUN

The full walkthrough

Recorded on the real setup: the phone logs in, scans the kiosk, and the session begins on its own.



THANK YOU

Thanks for the opportunity

This internship taught me a great deal. I got to work in C# and .NET, build a real desktop application with Avalonia, write a backend and web pages, package software for Linux, and integrate with hardware I did not control. All of it will help me going forward.

GUIDANCE

Eshwar Teja

For mentoring me through the project.

SUPPORT

Kalyan Sir

For his constant support and encouragement throughout.

Thank you — happy to take questions.